

UNIT 4

DATA AND ANALYSIS

10TH COMPUTER SCIENCE FEDERAL BOARD

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AGENDA

- Introduction
 - Data Science (DS)
 - Artificial Intelligence (AI) and its scope
 - ML Models
 - Machine Learning Process Steps
 - Churn Prediction and Behavioral Segmentations
 - Data Visualization
 - Database and ML
 - Stages of DS Life Cycle
 - Case Study
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DATA SCIENCE

- Data Science is the field of collecting, cleaning, and analyzing data to find useful patterns and make better decisions.
 - It combines math, statistics, programming, and domain knowledge.
 - The available data can be structured as well as unstructured.
 - Various programming languages like Python, R studio and MATLAB can be used for data visualization.
 - Real-life examples:
 - Netflix recommends movies based on your watch history
 - Amazon suggests products you might buy
 - Doctors predict diseases from patient data
 - Banks detect fraud transactions
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DATA SCIENCE AND ITS SCOPE



Healthcare: To predict the possibilities of some disease, based on X-Rays, Ultrasounds or other analytics of patient.



Sports: If we want to predict who is going to win a cricket match, based on the previous performance of teams, data science will help us.



Finance: In the banking and finance sector, data science is helpful in fraud detection. It is also helpful in developing trading and investment strategies.

DATA SCIENCE AND ITS SCOPE



Goods transportation: It helps in route optimization and minimize delivery times. Data driven approach helps to reduce cost. It also helps in real time tracking of transportation of goods.

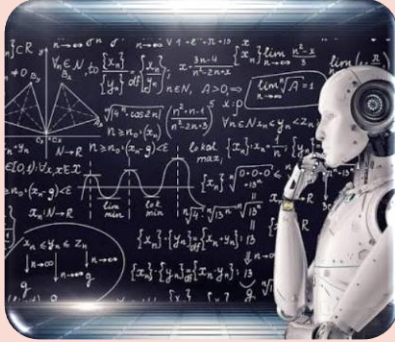


Airline: It helps in efficient route planning to reduce fuel consumption. It helps to improve services based on customer experience and feedback.



Energy and utilities: It helps to optimize energy production and distribution. Realtime monitoring is useful for load distribution and facilitates immediate response for demand fluctuations.

DATA SCIENCE AND ITS SCOPE



Education: It is useful to enhance personalized learning based on the performance of learners. It also helps to predict students' performance based on previous data.



Communication Media: It is used for content creation and distribution. Various approaches like sentiment analysis, social media monitoring and recommender systems are a few examples of scope of data science in media and communication sector.



Entertainment: Data science plays important role when you get suggestions about your favorite songs while playing online computer games.



Business: Data science helps in intelligent decision making about the business to get optimal profits. Customer insight and sentiments are also analyzed to improve the business.

ARTIFICIAL INTELLIGENCE (AI) AND ITS SCOPE

- The term artificial intelligence is not new. In 1950, a British mathematician Alan Turing proposed a Turing Test, which measured the ability of a machine to exhibit intelligent behavior. Nowadays, modern robots are supposed to be smarter if they can pass the Turing Test.
 - The term Artificial Intelligence refers to the ability of a machine to exhibit human behavior like problem solving, understanding natural language, and interacting with the environment intelligently.
 - we can discuss a few application of AI as:
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ARTIFICIAL INTELLIGENCE (AI) AND ITS SCOPE

AI Agents: Artificial intelligence provides bases to develop modern AI agents like Siri, Alexa, Google assistant, ChatGPT and Cortana.

Computer vision: It is a specialized branch of AI that teaches computers to draw meaningful results from digital images, videos and other visuals.

Healthcare: Like data science, artificial intelligence has played equally well in the field of healthcare. The examples are personalized treatment recommendations based on medical imaging.

Robotics: Whether you watch some latest robot, talking and cooking like a human being or you get a small robot to clean your house, all this technology is due to significant advancement of artificial intelligence.

Natural Language Processing (NLP): AI enabled the machines to understand and respond to the natural human languages. ChatGPT and chatbots are examples of it.
Now you can switch ON your air conditioner with voice command rather than picking up the remote controller.

ML MODELS: SUPERVISED LEARNING MODEL

- In this learning model a set of **labeled data** is provided to the machine/computer. The labeled data means that questions with their correct answers are provided to the machine. After reading/learning a lot of questions/answers the machine is trained. Now, only new questions without correct answers are provided to that machine and it provides us with the possible correct answers.
 - For example, if we provide the features (labels) of types of fruits i.e. shape, color and texture and give their names (answers) as well. After the machine is trained when we will give the features of a new fruit the supervised learning model will predict/provide the name of that fruit.
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ML MODELS: UNSUPERVISED LEARNING MODEL

- In unsupervised learning model, unlabeled data is provided to the machine/computer. The algorithm identifies the pattern in the data and divide it into groups based on similarities and differences. When the training is completed, the computer can identify new data and put it in the relevant category.
 - For example, if we provide various types of news to a computer then it will start dividing them into various groups, based on keywords. Suppose from a huge dataset the computer divides the news into various categories i.e. national politics, international politics, entertainment, medical, and technological news. When we give a new piece of information "Scientists successfully, saved the cancer patients at last stage" to the computer, it will categorize it into medical news.
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ML MODELS: REINFORCEMENT LEARNING MODEL

- In reinforcement learning the machines are trained with the help of positive or negative feedback. To explain reinforcement learning we will use the following key terms:
 - **Agent** (machine/computer)
 - **Environment** (where agent takes actions)
 - **Actions** (moves made by agent)
 - **Rewards** (points for helpful actions)
 - **Penalties** (points deducted for unhelpful actions)
 - The agent is provided with an environment where it must take certain actions, if the action is helpful towards the desired solution the agent will be rewarded with some points through feedback. If the actions are not helpful towards the desired solution the agent will be penalized (punished) by deducting the points. The agent improves its actions based on the feedback. This model is used to get optimal results.
 - Examples: Self-Driving Cars, AI Competition, Robots, Trading Algorithms
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MACHINE LEARNING PROCESS STEPS

Step 1: The preprocessed, cleaned data is split into **Training data** and **Testing data**.



Step 2: A suitable algorithm is selected to train the machine based on problem and data.



Step 3: The model is trained on the training set; it learns the pattern from data and develop a decision-making ability.



Step 4: After the model is trained, test data is provided to the model and results are compared with actual values. If needed, model is improved by changing the algorithm or by adjusting other necessary parameters.
